



mGURUSH-EQUITY B2C API Document



INTRODUCTION

The m-GURUSH API offers a framework and functionality for businesses that want to automate processes. The API is built using REST endpoints - it has predictable, resource-oriented URLs, and uses HTTP response codes to indicate API errors. JSON is returned by all API responses, including errors.

Base URLs

The base URLs for m-GURUSH API are

1. Development Environment
 - Protocol: HTTPS
 - Hostname:
sandbox.mgurush.com

The API is only available over HTTPS. Attempting to access the API over an unsecured HTTP connection will return an error.

AUTHENTICATION AND HEADERS

To make a REST API call, you must include request headers including the hmac header with an access_key.

Services listed in this API are invoked using HMAC Authentication. HMAC, or hashed message authentication code, is our preferred authentication scheme. It provides a straightforward but secure method of protecting your API key. It involves passing a hash composed of your api secret and the request. This hash is then signed with your api secret key. The result is a request that most importantly, does not contain your authentication key.

Header	Value
Content-Type	application/json
access_key	84902A4570346F1FEB05469C95F953DE9FCB985F
hmac	Hlkvwtc33oxKjG90xDDbWZyv7nSlzyE8iHryKh44zJQ=

API Keys

m-GURUSH wallet will generate a set of access key and secret keys for each partner.

Mime Formats

All requests and responses are JSON formatted. Providing the Accept header is optional, but recommended. The Content Type header must be provided when sending data to the API (using POST and PUT endpoints). You may pass either the standard JSON MIME type (application/json). You will receive an invalidContentType error if you attempt to make a POST/PUT request without one of these MIME types.



B2C TRANSACTION

Transaction B2C has two stage, one is validation of the customer with the mobile number and other one is processing the actual transaction. A transaction validation is recommended before posting an actual transaction. However, transaction posting can be done without validation request.

Validate customer:

Entity/Attribute	Mandatory	Description
mobileNumber	true	Mobile Number of Customer
amount	true	Transaction Amount
currency	true	Transaction Currency like SSP or USD

Validating a customer:

The request HTTP header:

```
POST      /bank/validateCustomer
Host:      sandbox.mgurush.com/b2b/api/v1/
Content-Type: application/json
access_key: 84902A4570346F1FEB05469C95F953DE9FCB985F
Hmac:      Hlkvwtc33oxKjG90xDDbWZyv7nSlzyE8iHryKh44zJQ
```

Example request body

```
{
  "mobileNumber": "333444000",
  "amount": "50000",
  "currency": "SSP",
}
```

Example response body

```
HTTP/1.1 200 OK
Content-type: application/json
{
  "status": {
    "statusCode": 0,
    "responseMessage": "Transaction Validation Receipt",
    "customerName": "Sanjeeb Bishoi",
    "amount": "50000",
    "currency": "SSP"
  }
}
```

Validation API will return the customer details to verify the receiver before sending the actual amount.



Post Transaction

Entity/Attribute	Mandatory	Description
mobileNumber	true	Mobile Number of Customer
amount	true	Transaction Amount
currency	true	Transaction Currency like SSP or USD
txnRefNumber	true	Unique transaction number from Bank
txnMode	true	Sender transaction mode like Cash, Cheque
description	false	Remarks of the transaction

Posting a transaction:

The request HTTP header:

```
POST      /bank/processTransaction
Host:     sandbox.mgurush.com/b2b/api/v1/
Content-Type: application/json
access_key: 84902A4570346F1FEB05469C95F953DE9FCB985F
Hmac:     Hlkvwtc33oxKjG90xDDbWZyv7nSlzyE8iHryKh44zJQ
```

Example request body

```
{
  "mobileNumber": "333444000",
  "amount": "50000",
  "currency": "SSP",
  "txnRefNumber": "8918371990",
  "txnMode": "Cash",
  "description": "To mGurush Wallet"
}
```

Example response body

```
HTTP/1.1 200 OK
Content-type: application/json
{
  "status": {
    "statusCode": 0,
    "responseMessage": "Transaction successful",
    "amount": "50000",
    "currency": "SSP",
    "customerName": "Sanjeeb Bishoi",
    "txnId": "8918371876"
  }
}
```

Bank can post both synchronous and asynchronous during a transaction.



Status Enquiry:

Example request body

```
{
  "txnRefNumber": "8918371990"
}
```

In response bank will get status of the transaction

Example response body

HTTP/1.1 200 OK

Content-type: application/json

```
{
  "status": {
    "statusCode": 0,
    "responseMessage": "Transaction successful",
    "amount": "50000",
    "currency": "SSP",
    "customerName": "Sanjeeb Bishoi",
    "txnId": "8918371876"
  }
}
```

Parsing response sent by m-GURUSH

Following fields will be sent as part of every response

Field Name	Type	Value
status	String	It will be either "Success" or "Failure"
statusCode	String	1 - Success, 0 - Failure

API Response Codes

You may encounter following response codes while making the api calls. Any unsuccessful response codes will contain more information to help you identify the cause of the problem.

Response Code	Meaning
200	OK. The request has succeeded.
201	Created. The request has been fulfilled and resulted in a new resource being created. The newly created resource can be referenced by the URI(s) returned in the entity of the response, with the most specific URI for the resource given by a Location header field.
204	No Content. The request was successful but no body was returned.
400	Bad Request. The request could not be understood by the server, usually due to malformed syntax. The response body will contain more details in the form of an array.



401	Unauthorized. The client has not provided a valid Authentication HTTP header or the user making the request has been disabled.
403	Forbidden. The client has provided a valid Authentication header, but does not have permission to access this resource.
404	Not Found. The requested resource was not found or the authenticated user cannot access the resource. The response body will explain which resource was not found.
405	Method Not Allowed. The HTTP verb used, or resource requested is not allowed. Note that we do not allow the PATCH verb to be used, and PUT should be used instead.
406	Not Acceptable. The content type specified in the Accept header was not acceptable to this endpoint.
409	Conflict. The resource to be created by your request already exists.
415	Unsupported Media Type. The response error will explain what types are accepted.
422	Unprocessable Entity. Could not process a POST or PUT request because the request is invalid. The response body will contain more details.
426	Upgrade Required. An unsecured connection was refused. Upgrade to TLS/SSL.
429	Too Many Requests. A rate limit has been reached. The headers will explain the details of the rate limit.
500	Internal Server Error. The server encountered an error while processing your request and failed. Please report this to the mGurush support team and quote the requestId.

APPENDIX

```
public void generateHmac(String payload) {  
  
    try {  
        Mac digest = Mac.getInstance("HmacSHA256");  
        SecretKeySpec secretKey = new SecretKeySpec("apiSecret", "HmacSHA256");  
        digest.init(secretKey);  
        final byte[] signatureBytes = digest.doFinal(payload);  
        byte[] encryptedByteArray = Base64.encodeBase64(signatureBytes);  
        logger.info("Hmac: " + new String(encryptedByteArray));  
        digest.reset();  
    }  
    catch (Exception e) { }  
}
```